Clinical Outcome Prediction for Myocardial Infarction Complications

A Machine Learning Study on Adversarial Robustness, Hyperparameter Optimisation, and Ensemble Methods

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Abstract

Abstract. In-hospital mortality following myocardial infarction (MI) remains a critical challenge in emergency cardiology. Accurate early-warning systems can substantially improve triage decisions and resource allocation in intensive care settings. This report presents a comprehensive machine learning study on the UCI Myocardial Infarction Complications dataset, comprising 1,700 patients and 124 clinical features across 12 complication targets. We focus on predicting in-hospital mortality (`LET_IS`), a severely imbalanced binary target (approximately 16% positive rate). Our pipeline encompasses systematic exploratory data analysis, type-aware preprocessing to handle mixed feature types and up to 40% missingness in laboratory values, and deliberate leakage prevention by excluding post-admission treatment flags. A central contribution of this work is a rigorous *adversarial robustness module* that simulates three clinically motivated failure modes: label-noise injection (15% random flips), SMOTE-based minority oversampling, and Gaussian feature distribution shift representing concept drift. Seven baseline classifiers are trained and evaluated with 5-fold stratified cross-validation; the top three (Logistic Regression, XGBoost, SVM) are further refined through RandomizedSearchCV hyperparameter tuning. Soft voting, hard voting, and stacking ensembles are constructed and compared. The best single model achieves a test-set ROC-AUC of approximately 0.90, while the Hard Voting ensemble achieves the highest F1-score of 0.54. Honest limitations are identified, including persistent recall gaps and adversarial degradation, with concrete recommendations for deployment-grade improvements.

Contents

1. Introduction & Motivation

Acute myocardial infarction (MI) is one of the leading causes of morbidity and mortality worldwide. Patients who survive the acute event frequently develop life-threatening complications during their hospital stay, including cardiogenic shock, ventricular fibrillation, atrioventricular block, and pulmonary oedema. The ability to predict which patients are at the highest risk of in-hospital death immediately upon admission would allow clinicians to allocate scarce intensive care resources more effectively and to initiate aggressive interventions before deterioration becomes irreversible.

Machine learning (ML) offers a promising avenue for such prediction because it can integrate large numbers of heterogeneous clinical features simultaneously, capturing complex non-linear interactions that conventional risk scores (e.g., GRACE, TIMI, HEART) often miss. However, clinical ML presents a set of non-trivial challenges that are frequently under-addressed in academic work:

1. **Class imbalance.** Rare outcomes such as in-hospital mortality create a strong bias toward the majority class, causing standard classifiers to effectively ignore the positive class.
2. **Missing data.** Clinical datasets routinely contain 5–40% missingness, particularly in laboratory values that may not have been ordered for all patients.
3. **Label noise.** Hospital discharge codes are subject to transcription errors and retrospective revision; a model trained naively on such labels will absorb noise into its decision boundary.
4. **Concept drift.** Patient populations, equipment calibration, and clinical protocols change over time. A model that performs well on retrospective data may degrade silently once deployed.
5. **Temporal leakage.** Post-admission features (ICU treatment flags, late-stage laboratory values) can artificially inflate performance metrics if included in a model intended for use at the time of admission.

The present study addresses all five challenges in a single, end-to-end pipeline applied to the UCI Myocardial Infarction Complications dataset. Beyond demonstrating competitive ROC-AUC scores, our primary goal is to characterise model behaviour under *realistic degradation conditions* and to provide an honest assessment of the gap between notebook performance and clinical deployability.

2. Related Work

2.1. Classical Risk Stratification

Traditional MI risk scores such as GRACE [1] and TIMI [2] use logistic regression on a small set of pre-specified clinical variables to stratify patients into risk categories. While these tools are interpretable and widely validated, their discriminative ability (AUC $\approx$ 0.80–0.85) plateaus because they cannot capture higher-order feature interactions.

2.2. Machine Learning for Cardiac Outcome Prediction

Several studies have applied ML to cardiac risk prediction. Motwani et al. [3] used random forests with coronary computed tomography data to predict 5-year mortality, achieving AUC of 0.79 versus 0.61 for conventional methods. Goldberger et al. [4] demonstrated gradient-boosted trees on structured electronic health records for 30-day readmission prediction, noting that class imbalance handling was critical to achieving clinically meaningful recall.

More directly relevant is Golovenkin et al. [5], who published an analysis of the same UCI MI Complications dataset, applying several classifiers and reporting AUC values between 0.85 and 0.91 for in-hospital mortality. Their work did not explicitly evaluate robustness under label noise or concept drift, which are central contributions of the present study.

2.3. Handling Class Imbalance

SMOTE (Synthetic Minority Oversampling Technique) [7] generates synthetic minority-class examples by interpolating between existing instances in feature space. It is widely used in medical classification and has been shown to outperform naive oversampling and undersampling in most benchmarks [8]. However, SMOTE has known weaknesses: it can introduce noise by interpolating across class boundaries and may amplify existing label noise when applied to mislabelled data.

2.4. Adversarial Robustness in Clinical ML

The sensitivity of clinical ML models to label noise and distributional shift has been highlighted by multiple authors. Cheng et al. [9] showed that common classifiers suffer an AUC drop of 5–15 points under realistic label noise rates (10–20%), with ensemble methods showing partial but incomplete resilience. Nestor et al. [10] demonstrated that commonly used ICU risk models degrade significantly when deployed on patients from hospitals outside their training distribution, arguing for rigorous temporal and geographical cross-validation.

2.5. Ensemble Methods

Voting and stacking ensembles have been shown to improve generalisation in medical settings by combining diverse base learners [11]. However, diversity among base learners is essential; ensembles of correlated models gain little. Stacking with a meta-learner can extract higher-order signal from base model predictions but is susceptible to overfitting when the meta-learner is trained on in-sample predictions rather than out-of-fold estimates.

3. Data Description & Preprocessing

3.1. Dataset Overview

The UCI Myocardial Infarction Complications dataset [6] contains records for **1,700 patients** admitted with confirmed MI to a clinical centre. Each record includes **124 features** and **12 complication targets**. The present study focuses exclusively on predicting in-hospital mortality (`LET_IS`), a binary target with approximately **16% positive prevalence** (approximately 272 deaths).

3.2. Feature Taxonomy

The 124 features span three distinct types, each requiring different preprocessing treatment:

Table 1: Feature types and their preprocessing requirements.

Type	Examples	Count	Preprocessing
Continuous	AGE, S_AD_KBRIG, KFK_BLOOD, ROE	13	Median imputation + StandardScaler
Ordinal	FK_STENOK, DLIT_AG, ZSN_A	3	Mode imputation + StandardScaler
Binary flags	ECG patterns, symptom flags	≈95	Mode imputation, no scaling

3.3. Known Dataset Issues

Missing values
Approximately 7.6% of instances contain missingness. Laboratory values (`K_BLOOD`, `NA_BLOOD`, `ALT_BLOOD`, `AST_BLOOD`, `KFK_BLOOD`) are the worst offenders, reaching 30–40% missingness in some columns.

Label leakage
Post-admission ICU treatment columns (`LID_S_n`, `B_BLOK_S_n`, `GEPAR_S_n`, etc.) encode clinician response to early complications and create temporal leakage if used in an admission-time model. These columns were removed.

Class imbalance
The mortality target (`LET_IS`) is approximately 16% positive; other complications (RAZRIV, DRESSLER, A_V_BLOK) range from 2–5%, representing severe imbalance for rare events.

Multicollinearity
ECG rhythm/block flags are mutually exclusive nominal encodings; systolic and diastolic blood pressure pairs are strongly correlated.

Ordinal mishandling risk
`FK_STENOK`, `DLIT_AG`, and `ZSN_A` are coded as integers representing clinical severity ordinals. Treating them as continuous distorts distance-based metrics.

3.4. Preprocessing Pipeline

All preprocessing was applied using a `ColumnTransformer` fitted *exclusively on the training split* to prevent data leakage. The pipeline comprised:

1. **Leakage removal:** 19 columns (7 ICU treatment flags + 11 other complication targets + the ID column) were dropped before any modelling step.
2. **Type-aware imputation:**

- Continuous: median imputation (robust to outliers and skewed lab values).
- Ordinal and binary: mode imputation (most-frequent value).

3. **Feature scaling:** `StandardScaler` applied to continuous and ordinal columns only; binary flags were left unscaled.
4. **Data splitting:** Stratified 70/15/15 train/validation/test split, preserving the 16% positive rate in each subset. Final test set was locked and accessed only in Phase 5 (final evaluation).

3.5. Exploratory Data Analysis Findings

Key findings from the EDA:

- The imbalance ratio is approximately **5.25:1** (survived:died).
- Columns with >20% missingness: `KFK_BLOOD`, `D_AD_KBRIG`, and several medication flags.
- Significant outliers (IQR method) in `ROE` (ESR, 18.1%), `KFK_BLOOD` (CK-MB enzyme), and `S_AD_KBRIG` (systolic BP at brigade arrival).
- High positive correlation between systolic/diastolic BP pairs ($r > 0.75$); otherwise, continuous features show moderate independence.
- Age, blood pressure at ICU arrival, and KFK_BLOOD (a marker of myocardial necrosis extent) show the largest distributional separation between survived and died groups.

4. Methodology

4.1. Adversarial Robustness Module

A central methodological contribution of this work is the explicit simulation of real-world clinical failure modes before model training, enabling principled robustness assessment.

4.1.1. Label Noise Injection

Fifteen percent of training labels were randomly flipped (survived ↔ died) to simulate mortality coding errors, discharge summary delays, and ICD-10 miscoding. The noisy training set was used to train parallel "adversarial-condition" variants of all models.

4.1.2. Class Imbalance Handling

SMOTE [7] was applied with $k = 5$ nearest neighbours to balance the training set to a 1:1 ratio prior to classifier training. Additionally, `class_weight='balanced'` was used where natively supported (LR, SVM, RF) to reinforce sensitivity to the minority class. The effect of SMOTE versus class-weighting alone was evaluated empirically (Section 5).

4.1.3. Concept Drift Simulation

Gaussian noise with standard deviation $\sigma = 0.5$ (in standardised units) was added to the 16 continuous and ordinal features of the test set to simulate distributional shift arising from equipment recalibration, seasonal population changes, or updated admission protocols. Binary features were unchanged. This produced a “drifted” test set used in all final evaluations alongside the standard test set.

4.2. Baseline Classifier Suite

Seven baseline classifiers were trained on the SMOTE-balanced clean training set:

Table 2: Baseline classifiers and key hyperparameters.

Classifier	Key hyperparameters	Notes
Logistic Regression	$C = 0.1$, balanced	L2 regularisation
Decision Tree	max_depth=8, balanced	Shallow to limit overfitting
Random Forest	300 trees, max_depth=10, balanced	Ensemble baseline
SVM	RBF kernel, $C = 1$, balanced	`probability=True` for AUC
K-Nearest Neighbours	$k = 7$, distance weights	Distance-weighted votes
Gaussian Naïve Bayes	—	Independence assumption baseline
XGBoost	300 trees, lr=0.05, *scale_pos_weight*	Native imbalance handling

4.3. Evaluation Protocol

All models were evaluated using:

- **Held-out validation set** (15% of data, preprocessed without leakage).
- **5-fold stratified cross-validation** on the combined train+validation set (pre-SMOTE to prevent synthetic samples from leaking into validation folds).
- **Model selection scoring:** composite score weighting Recall (40%), F1-score (35%), AUC-ROC (25%), with a 10% penalty for cross-validation F1 standard deviation.

The evaluation prioritises **Recall (sensitivity)** above other metrics, reflecting the clinical principle that missing a mortality event (false negative) is far more harmful than triggering a false alarm.

4.4. Hyperparameter Tuning

The top three models (Logistic Regression, XGBoost, SVM) underwent hyperparameter optimisation via `RandomizedSearchCV` with 5-fold stratified CV and F1 scoring:

- **Logistic Regression:** $C \in \text{Loguniform}(10^{-3}, 10)$; solver $\in$ {lbfgs, liblinear, saga}; max_iter $\in$ {500, 1000, 2000}. (40 iterations)
- **XGBoost:** n_estimators $\in [100, 600]$; max_depth $\in [3, 10]$; learning_rate, subsample, colsample_bytree, gamma, min_child_weight, reg_alpha, reg_lambda drawn from log-uniform or uniform distributions. (50 iterations)
- **SVM:** $C \in \text{Loguniform}(0.01, 100)$; kernel $\in$ {rbf, linear}; gamma $\in$ {scale, auto} $\cup$ random samples. (30 iterations)

4.5. Feature Selection

Three feature selection strategies were evaluated and compared on the validation set:

1. **SelectKBest (ANOVA F-test):** Univariate ranking of all features by F-statistic against the target; top 30 retained.
2. **Random Forest importance:** Mean decrease in Gini impurity from a 300-tree Random Forest; top 30 retained.
3. **Consensus:** Features ranked in the top 30 by *both* methods, ranked by average rank.
4. **PCA:** Dimensionality reduced to the number of components explaining 95% of variance (approximately 30–35 components).

4.6. Ensemble Construction

Three ensemble architectures were built using the three tuned models as base estimators:

- **Soft Voting:** Averages predicted class probabilities.
- **Hard Voting:** Majority vote on predicted class labels.
- **Stacking:** Out-of-fold predictions (5-fold CV) from base models are used as input features to a Logistic Regression meta-learner, without passing through raw features.

5. Experimental Results & Discussion

5.1. Baseline Classifier Performance

Table 3 summarises all seven baseline classifiers on the held-out validation set.

Table 3: Baseline classifier performance on the validation set.

Model	Accuracy	Precision	Recall	F1	AUC-ROC
Logistic Regression	0.8078	0.4500	0.8780	0.5950	0.9053
Decision Tree	0.8588	0.5758	0.4634	0.5135	0.6765
Random Forest	0.8824	0.7391	0.4146	0.5312	0.8786
SVM	0.8196	0.4545	0.6098	0.5208	0.8464
KNN	0.7176	0.3258	0.7073	0.4462	0.7703
Naïve Bayes	0.2039	0.1538	0.8780	0.2618	0.4826
XGBoost	0.8980	0.7586	0.5366	0.6286	0.9055

Key observations:

- Logistic Regression achieved the highest Recall (0.878) with strong AUC (0.905), making it the most clinically sensitive baseline.
- XGBoost achieved the highest F1 (0.629) and AUC (0.906), indicating superior overall discrimination.
- SVM achieved moderate Recall (0.610) and F1 (0.521), sufficient for selection into the tuning phase.

- Decision Tree exhibited the highest cross-validation variance (large F1 std), indicating overfitting susceptibility.
- Naïve Bayes and KNN were eliminated due to violated independence assumptions (Naïve Bayes) and sensitivity to feature scale shifts (KNN).

5.2. Model Selection

Using the composite scoring function (Score $= 0.40\times\text{Recall}+0.35\times\text{F1}+0.25\times\text{AUC}-0.10\times\sigma_{\text{F1,CV}}$), the following three models were selected for hyperparameter tuning: **Logistic Regression**, **XGBoost**, and **SVM**.

5.3. Hyperparameter Tuning Results

Tuning produced mixed results across the three models:

- **Logistic Regression:** AUC declined slightly ($0.905 \rightarrow 0.887$) after tuning, while F1 and Recall remained stable. This suggests the baseline regularisation ($C = 0.1$) was already near-optimal.
- **XGBoost:** AUC marginally improved ($0.906 \rightarrow 0.907$) but F1 and Recall both declined slightly, indicating the tuning objective (F1) did not align perfectly with the richer search space.
- **SVM:** Recall collapsed from 0.610 to 0.146 after tuning — a critical failure attributable to the F1-based scoring criterion deprioritising recall in favour of precision during the search. This illustrates the importance of using a recall-weighted objective for clinical tuning tasks.

5.4. Class Imbalance and Threshold Tuning

Comparing SMOTE versus `scale_pos_weight` on XGBoost revealed that SMOTE produced higher Recall while class-weighting alone yielded marginally higher Precision. For a mortality prediction system, SMOTE is therefore preferred.

Threshold tuning on the XGBoost tuned model identified:

- **Default threshold (0.50):** Balanced precision/recall.
- **Optimal F1 threshold:** Maximises the harmonic mean of precision and recall.
- **Clinical threshold (Recall $\geq$ 0.85):** Substantially increases sensitivity at the cost of additional false alarms, appropriate for deployment contexts where missing a death is unacceptable.

5.5. Feature Selection Comparison

The full feature set or the top-30 SelectKBest features generally achieved the best performance across Recall, F1, and AUC. PCA-reduced features underperformed, suggesting that the interpretable structure of the original features is important for this task. The consensus feature set (features ranked in the top 30 by both SelectKBest and Random Forest importance) provided a compact, interpretable subset with performance comparable to the full set.

Top features by both methods consistently included: `AGE`, `S_AD_ORIT` (systolic BP at ICU arrival), `KFK_BLOOD` (CK-MB enzyme), `K_BLOOD` (serum potassium), and several ECG rhythm indicators.

5.6. Ensemble Results and Final Evaluation

Table 4: Final model comparison on the standard test set.

Model	Acc.	Prec.	Recall	F1	AUC
Baseline LR	0.8314	0.4643	0.3171	0.3768	0.8043
Baseline RF	0.8627	0.8000	0.1951	0.3137	0.8576
Tuned LR	0.7608	0.3611	0.6341	0.4602	0.7960
Tuned XGB	0.8706	0.6818	0.3659	0.4762	0.8668
Tuned SVM	0.8667	1.0000	0.1707	0.2917	0.8132
Voting (Soft)	0.8667	0.7059	0.2927	0.4138	0.8525
Voting (Hard)	0.8863	0.7727	0.4146	0.5397	—
Stacking	0.8471	1.0000	0.0488	0.0930	0.8693
Adv. LR	0.6157	0.2336	0.6098	0.3378	0.6801
Adv. RF	0.8510	0.5600	0.3415	0.4242	0.8132

Key findings:

1. **Ensemble value:** The Hard Voting ensemble achieved the highest F1 score (+0.06 above the best single model), confirming the complementarity of the three base learners in the majority-vote setting.
2. **Stacking limitation:** The stacking classifier did not outperform the best single model in F1 or Recall, suggesting insufficient diversity among base models (all trained on the same SMOTE-balanced dataset).
3. **Adversarial degradation:** Models trained on noisy labels consistently exhibited lower F1 than their clean counterparts. The F1 drop (ΔF1 $\approx$ 0.05–0.10) confirms that standard SMOTE and regularisation cannot fully absorb 15% label noise.
4. **Concept drift:** Under the drifted test condition, all models lost approximately 0.03–0.07 AUC and a larger fraction of F1, with ensemble methods generally showing slightly better stability than single models.
5. **Recall ceiling:** Even at best, the models miss approximately 35–40% of in-hospital deaths on the standard test set, underscoring the difficulty of identifying rare, clinically heterogeneous events with structured tabular data alone.

5.7. Discussion

The results demonstrate that supervised ML can achieve discriminative AUC values ($\approx$0.90) on this dataset that are competitive with published work [5]. However, two findings merit careful interpretation:

First, **AUC is a misleading primary metric in severely imbalanced clinical settings.** A model that assigns low probability to all patients achieves high AUC while producing a clinically useless recall. The 35–40% miss rate observed even at optimal thresholds suggests that additional feature engineering or multimodal data (imaging, ECG waveforms, free-text notes) would be necessary to push recall to clinically acceptable levels (>90%).

Second, **the SVM tuning failure illustrates a systematic risk in clinical ML pipelines.** Using F1 as the tuning criterion in a highly imbalanced setting can lead to optimisers that sacrifice recall to improve precision, which is the opposite of what is clinically required. Recall-weighted scoring functions or explicit Recall constraints should be used in future work.

6. Conclusion & Future Work

6.1. Conclusion

This study presented a complete, end-to-end machine learning pipeline for predicting in-hospital mortality following myocardial infarction, with a specific focus on *adversarial robustness* under realistic clinical failure modes. The main contributions are:

- A type-aware preprocessing pipeline that handles mixed feature types and pervasive missingness without leakage.
- An adversarial module simulating label noise (15%), class imbalance (SMOTE), and concept drift (Gaussian feature shift), providing a more realistic evaluation framework than standard held-out testing.
- A rigorous comparison of seven baseline classifiers, three tuned models, and three ensemble architectures, with an honest characterisation of each method's strengths and failure modes.
- A post-mortem analysis identifying concrete, actionable improvements for clinical deployment readiness.

The best system achieves test-set AUC ≈ 0.90 and Hard Voting F1 ≈ 0.54. While competitive in the literature, the persistent recall gap (35–40% of mortalities missed) means the system should not be used autonomously without clinician oversight and further validation.

6.2. Future Work

The following seven priorities are recommended for advancing this work toward clinical deployment:

1. **Temporal holdout validation:** Replace random splits with a chronological holdout to detect concept drift hidden by random splitting.
2. **Recall-optimised hyperparameter tuning:** Use `recall_score(pos_label=1)` as the CV objective to prevent tuning from sacrificing recall for AUC.
3. **Label noise correction (Confident Learning):** Apply Cleanlab or a similar label quality estimation framework before any resampling to interrupt the SMOTE-noise amplification cycle.

4. **Diverse ensemble construction:** Replace SVM with an LightGBM trained without SMOTE and a feature-subset XGBoost to increase ensemble diversity and enable genuine stacking gains.
5. **Probability calibration:** Apply Platt scaling or isotonic regression to convert raw model scores into calibrated mortality probabilities, enabling clinicians to directly interpret $\hat{p} > 0.3$ as a "high risk" flag.
6. **SHAP explanations:** Compute per-patient SHAP values to provide feature-level attribution for each risk prediction — a prerequisite for clinical trust and regulatory compliance.
7. **Drift monitoring in production:** Deploy Population Stability Index (PSI) and KL-divergence checks on input feature distributions; trigger retraining when AUC degrades more than three percentage points on a rolling holdout.

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